

Data Analysis Techniques for Fraud Examiners



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Course Objective

The objective of this course is to introduce fraud examiners to the basics of using data analysis techniques to uncover fraud. You will work through several interactive examples to illustrate simple analysis techniques. With these examples, you will learn how to perform a variety of tests and interpret the results to identify common red flags of fraud.

Course Objective

Keep in mind that fraud examiners use a seemingly limitless number of data analysis techniques. The techniques you employ will be unique to each investigation you work on. However, this course will introduce you to some commonly used techniques that are applicable to many types of investigations. Once you learn these techniques, you can figure out on your own the appropriate analysis to conduct for your particular investigation.

Introduction

Another point to keep in mind is that data analytics alone is typically not a sufficient means of detecting fraud. Rather, the objective of using data analytics is to narrow the scope of an investigation. If data analysis turns up anomalies and red flags, the fraud examiner will know where specifically to focus his investigation.

Course Overview

This course is organized into three parts made up of 22 lessons:

Part I—Data Analytics

- **Lesson 1:** What Is Data Analytics?
- **Lesson 2:** Why Do We Use Data Analytics to Detect Fraud?
- **Lesson 3:** Advantages of Using Data Analytics to Detect Fraud
- **Lesson 4:** Challenges of Using Data Analytics to Detect Fraud

Part II—The Data Analysis Process

- **Lesson 5:** The Planning Phase
- **Lesson 6:** The Preparation Phase
- **Lesson 7:** The Testing and Interpretation Phase
- **Lesson 8:** The Post-Analysis Phase

Course Overview

Part III—Data Analysis Techniques

- **Lesson 9:** Basic Analytical Procedures
- **Lesson 10:** Data Analytics and Scheme-Specific Frauds
- **Lesson 11:** Billing Schemes
- **Lesson 12:** Check Tampering Schemes
- **Lesson 13:** Payroll Schemes
- **Lesson 14:** Expense Reimbursement Schemes
- **Lesson 15:** Theft of Cash Receipts Schemes
- **Lesson 16:** Noncash Asset Schemes
- **Lesson 17:** Corruption Schemes
- **Lesson 18:** Financial Statement Fraud Schemes
- **Lesson 19:** Financial Statement Analysis
- **Lesson 20:** Benford's Law
- **Lesson 21:** Textual Analytics
- **Lesson 22:** Conclusion

This course will teach students how to:

- Recall the benefits and challenges of using data analytics.
- Identify the four phases of the data analysis process.
- Recognize common red flags for data that appears to be suspicious or fraudulent.
- Recognize various data analysis techniques that can be applied based on the type of fraud scheme(s) under investigation.

ACFE ONLINE SELF-STUDY CPE COURSE

Part I: Data Analytics



Introduction

Data analysis is a powerful tool for fraud examiners. By searching for patterns, anomalies, trends, and outliers, a fraud examiner might discover a very cleverly hidden scheme. A good fraudster is adept at covering his tracks. However, data analytics empowers an investigator to uncover even the most thoroughly shrouded fraud. Furthermore, data analytics can be used to continuously monitor an organization so that red flags of fraud are discovered and investigated as soon as they occur.

Lesson 1: What Is Data Analytics?



Introduction

Data analysis identifies trends, patterns, anomalies, and exceptions within data. There are several data analysis techniques available to fraud examiners that help detect red flags and address areas that pose the greatest risks. Data analysis techniques are especially useful when trying to find fraud that is hidden in large amounts of data and manual checks are insufficient.

Defining Data Analysis

Vincent Walden, CFE, CPA, is a partner at Ernst & Young in the Fraud Investigation and Dispute Services division. Listen as he defines data analysis in his own words.



Big Data

Big data is a buzzword in the business, audit, and fraud investigation realms. According to the technology research firm Gartner, *big data* refers to high volume, velocity, and variety information assets that demand cost-effective, innovative forms of information processing for enhanced insight and decision making.

Big data does not comprise a single set of data; rather, it looks at the way data grow when you consider methods of connecting different data sets together to create even more sets of information.

Big Data

Data is either structured or unstructured. *Structured data* is the type of data that you would find in a database, consisting of recognizable and predictable structures. Examples of structured data include sales records, payment or expense details, and financial reports.

Unstructured data, by contrast, is data that you would not find in a traditional spreadsheet or database. Examples of unstructured data include vendor invoices, email, user documents, human resources files, social media activity, corporate document repositories, and news feeds.

Applying Fraud Knowledge

When using data analysis to conduct a fraud examination, the fraud examiner might use structured data, unstructured data, or a combination of the two. For example, conducting an analysis on email correspondence (unstructured data) among employees might turn up suspicious activity in the purchasing department. Upon closer inspection of the inventory records (structured data), the fraud examiner might uncover that an employee has been stealing inventory and covering his tracks in the records.

Applying Fraud Knowledge

Performing data analysis techniques, including the necessary data extraction and cleansing, requires a combination of fraud investigation skills and technological savvy, in addition to relevant industry knowledge. Fraud examiners must be able to translate their knowledge of their organization and their understanding of common fraud indicators into analytics tests to help detect fraud.

Effective Approaches to Data Analysis

To conduct effective data analysis, fraud examiners must take a wide-ranging approach. The more creative fraudsters get in hiding their schemes, the more creative the fraud examiner must become in analyzing data to detect these schemes.

Effective Approaches to Data Analysis

Data analysis alone is unlikely to detect fraud. Rather, data analysis is used to quickly and thoroughly examine large volumes of data to find anomalies. Once these anomalies are identified, the fraud examiner can focus his attention on them to determine whether fraud is present. If fraud can be thought of as “a needle in a haystack,” data analytics is the tool that points where in the haystack to look for it.

Lesson 2: Why Do We Use Data Analytics to Detect Fraud?



Introduction

Some businesses have millions of transactions. This vast amount of data makes it easier for a fraudster to conceal his wrongdoing while also making it more difficult for an investigator to detect any misconduct. The process of manually reviewing all documents is not an efficient use of time or resources. This is why data analytics is used—to examine millions of records quickly, easily, and efficiently so that fraud examiners can narrow their scope and focus their investigation on suspicious data.

Why Use Data Analytics?

Data extraction and analysis software assists a fraud examiner by highlighting individual transactions that have characteristics often associated with fraudulent activity. The capabilities of data analytics software are nearly boundless. The software affords the fraud examiner the ability to analyze data from previous years, compare data from different locations, and perform continuous monitoring to identify anomalies as they pop up.

Why Use Data Analytics?

According to the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, more than 39 percent of frauds reported were detected by whistleblowers or through tips. That means close to 60 percent of frauds were detected by other means. Data analysis could be used to detect these frauds before they are even noticed and reported by a whistleblower.

Who Uses Data Analytics?

Data analysis is a relevant tool for any fraud examiner. The reality of the modern business world is that companies have the ability to house and process large volumes of transactional data. Technology presents unique opportunities for both fraudsters and fraud investigators. As fraudsters become more adept at hiding their wrongful activity, fraud examiners must be able to use the technological tools available to uncover even the most well-hidden schemes.

Who Uses Data Analytics?

Data analysis has long been a tool in the auditor's toolbox. However, really anyone with access to large amounts of data—external auditors, internal auditors, accountants, fraud investigators, law enforcement personnel, litigation support—is likely to utilize the benefits of data analysis.

Data Analytics and Fraud



Trends in recent legislation and business practice have put a high importance on the need to promptly and efficiently investigate allegations and proactively address fraud risk. Data analytics affords auditors and investigators the ability to do this.

Data Analytics and Fraud

Data analytics is useful in both the prevention and detection of fraud. To prevent fraud, organizations can employ repetitive or continuous data analysis by creating scripts to run against large volumes of data to identify outliers as they occur over time. By investigating these anomalies, the organization might be able to strengthen controls or even stop a potential fraud in its tracks before it becomes destructive. In addition, if employees know that every transaction running through the systems is monitored, they are less inclined to engage in wrongful behavior.

Lesson 3: Advantages of Using Data Analytics to Detect Fraud



Introduction

Organizations can gain confidence in the results of data analysis through the automation of data analysis tools and the sophistication of software. One of the greatest advantages of using data analytics to detect fraud is the increased efficiency, effectiveness, and accuracy of these techniques. With data analytics, fraud examiners can examine large data sets in a much shorter period of time than could otherwise be accomplished.

Advantages of Data Analytics

Data analytics makes possible the testing of up to 100 percent of relevant data and transactions across multiple applications and systems. When considering the data population, fraud examiners should determine how much historical data is needed to have a significant impact on the results. If the fraud examiner only tests partial data, he might not detect control breaches and suspicious transactions. The impact of control failures might not be fully quantified and anomalies might go undetected.

Advantages of Data Analytics

Brenda Buetow, CFE, has experience in data mining, forensic accounting, and investigative services. In this clip, she discusses why data analysis is helpful for a fraud investigator.

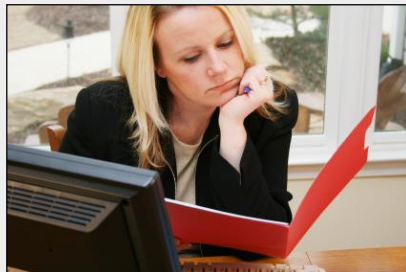


Benefits of Using Data Analytics

In summary, the benefits of using data analytics for fraud prevention and detection include the ability to:

- Proactively identify items that appear to be fraudulent.
- Focus fraud detection efforts on suspicious transactions.
- Determine if a company's internal controls are working as intended.
- Analyze business transactions at the source level, allowing for a more complete view.

Lesson 4: Challenges of Using Data Analytics to Detect Fraud



Introduction

As effective a tool as data analytics can be, it is not without its challenges to a fraud examiner. Some of the greatest challenges of using data analytics to detect fraud include:

- Poorly defined scope
- Problems obtaining the data
- Manually maintained data
- False positives
- Failing to ensure the integrity of the data
- Improper extraction
- Misinterpretation of the data
- Failing to consider reality

Challenges of Using Data Analytics

Poorly Defined Scope

When planning an investigation using data analytics, the fraud examiner must define the scope. A poorly defined scope can lead to inaccurate results. Fraud examiners should first consider the purpose of the investigation and then determine what data might be needed. It is also important for fraud examiners to consider what tests might be useful to their investigation so that results are valuable and relevant.

Challenges of Using Data Analytics

Problems Obtaining the Data

One of the ultimate challenges in using data analytics to detect fraud is acquiring the necessary data on which to perform the tests. The quality of the data obtained affects the results of analytics tests for better or worse. However, obtaining the data can be a slow process—sometimes met with resistance by internal parties or the client—and the results do not always reflect the requested data. The data needed might also be housed in outdated systems or even maintained manually, making it all the more difficult to acquire.

Challenges of Using Data Analytics

Manually Maintained Data

It should not be assumed that all data is housed electronically. Even though most data is stored electronically, that might not be the case for some organizations. Fraud examiners should consider what data is manually maintained and how relevant it is to their investigation. Data that is manually maintained is prone to issues related to data integrity. If possible, automated data should be used when performing data analysis.

Challenges of Using Data Analytics

False Positives

Reducing the number of false positives is a challenge faced by fraud examiners when designing data analysis techniques. A large number of false positives can slow down an investigation because a fraud examiner has to sort through and separate the true exceptions from the ones that are false. Quality assurance tests run on data during the design process can help eliminate a large number of false positives. Taking the time to perform assurance testing can save fraud examiners time and effort in the long run.

Challenges of Using Data Analytics

Failing to Ensure the Integrity of the Data

In any investigation, fraud examiners should be cautious of the evidence with which they are working. They should ensure that it is complete, valid, and that it has not been altered. A data analysis engagement is no different. Data integrity can make or break the results of the investigation. Fraud examiners should assess the integrity of the data they gather before performing analysis.

Challenges of Using Data Analytics

Failing to Ensure the Integrity of the Data

When fraud is suspected, the integrity of the data and the accuracy of the analysis are central to a successful investigation. If an investigator makes a wrongful accusation, it can seriously damage the reputation of the accused and also create a potential legal liability for the investigator. Fraud examiners must assess the consequences of relying on faulty results and determine the amount of testing and verification necessary to reduce those consequences to an acceptable level. If this is not possible with a particular data source, fraud examiners might need to remove that source from the investigation.

Challenges of Using Data Analytics

Improper Extraction

There are many ways in which an auditor who relies too much on abstract analysis can fail to arrive at correct conclusions. For example, auditors must avoid improperly extracting the source data. This caveat relates back to data integrity, but is not entirely the same. A fraud examiner might have perfectly clean and complete data to work with, but, if data are transformed when extracted, the information might be subject to erroneous analysis. Fields can be lost, decimal places shifted, or data corrupted. Extracted data must be compared to original data using control totals and other methods to ensure reliability.

Challenges of Using Data Analytics

Misinterpreting the Data

Even if data are correct and complete, the fraud examiner can misread them. Misreading the data puts the fraud examiner at risk of jumping to incorrect conclusions, which could result in wrongful accusations. For example, a file containing both debits and credits might be read as debits only. Moreover, a file containing dollar amounts might be interpreted using the wrong currency or an old currency conversion date.

Challenges of Using Data Analytics

Failing to Consider Reality

Even with uncorrupted data that are correctly understood, practical factors can make a great difference in a fraud examiner's conclusions. This is why it is vital to thoroughly investigate any red flags that arise before drawing conclusions. In one case, an auditor found many anomalies in an organization's data during the month of June. Upon further investigation, he learned that the controller had taken a vacation during that time. In his absence, a temporary employee was hired to perform some of his simpler tasks. The temp made many errors during this time, but the mistakes did not necessarily indicate that the temp committed fraud.

Other Challenges

Other potential challenges the fraud examiner might face using data analytics include:

- Lack of familiarity with data storage or software systems
- Lack of familiarity with organizational processes surrounding data
- Source data completeness
- Privacy and confidentiality concerns

Other Challenges

- Cost of data analytics software
- Continuous evolution of fraud schemes to bypass existing detection techniques
- Element of concealment inherent in frauds
- Learning curve associated with data analytics techniques and identification of appropriate methods to use in a particular case

Part II: The Data Analysis Process



Introduction

Although the crux of data analysis involves running targeted tests on data to identify anomalies, the ability of such tests to help detect fraud depends greatly on what the fraud examiner does before and after actually performing the data analysis techniques. To ensure the most accurate and meaningful results, a formal process should be applied that includes four phases:

1. The planning phase
2. The preparation phase
3. The testing and interpretation phase
4. The post-analysis phase

Lesson 5: The Planning Phase



Introduction

As with any process, proper planning is essential. When planning a data analysis engagement, the fraud examiner should devote sufficient time and attention to planning. If he does not perform effective planning, he runs the risk of analyzing irrelevant data, losing focus on the purpose of the engagement, running into technical difficulties that could have been avoided, and overlooking transactions or areas that might pose significant risk to the organization.

The Planning Phase

The planning phase of a data analysis engagement includes the following steps:

- Step 1: Understand the data.
- Step 2: Articulate examination objectives.
- Step 3: Build a profile of potential frauds.
- Step 4: Determine whether predication exists.

The Planning Phase

Step 1: Understand the Data

As a first step—long before determining which tests to run—the fraud examiner must determine what data are available. The fraud examiner should begin by identifying the data sources, file formats, and key systems used. He might have to consult with the data administrator or accounting manager to learn what fields and records exist, and in which tables the information is stored. He also needs to know how any tables might be linked together. It is also important for him to get the names of business process owners in areas that he intends to examine.

The Planning Phase

Step 1: Understand the Data

A fraud examiner should keep in mind that gathering relevant data is crucial to the outcome of the examination. He should understand what data sources might prove useful. Examples include:

- Accounting and financial data
- Human resources data
- Customer data
- Vendor data
- Internal communications and documents
- External benchmarking data

The Planning Phase

Step 1: Understand the Data

Understanding the structure of the existing data not only helps ensure that a fraud examiner builds workable tests to be run on the data, but might also help identify additional areas for exploration.



The Planning Phase

Step 2: Articulate Examination Objectives

Many companies process millions of transactions every day. Even small organizations might store huge volumes of data that need to be considered for analysis. Going through every single piece of data manually would be both impossible and inefficient. Data analysis eliminates the need for investigators to perform this onerous task. However, conducting every possible analysis of the data would be prohibitively time-consuming as well. For this reason, a fraud examiner should articulate his examination objectives as soon as he understands what data are available.

The Planning Phase

Step 2: Articulate Examination Objectives

When articulating the objectives of a data analysis engagement, fraud examiners should consider:

- The impetus for the engagement
- The structure and size of the business
- The target area of the examination
- The resources (e.g., time, personnel, etc.) available
- Whether any predication exists that a particular fraud is occurring
- Any there any known frauds that were considered material in nature

The Planning Phase

Step 3: Build a Profile of Potential Frauds

Any data analysis performed should be based on the entity's existing fraud risks. This not only saves time in the engagement, but significantly increases the chances that the analysis will generate useful results. To start, build a profile of potential frauds by identifying the organization's risk areas, the types of frauds possible in those risk areas, and the resulting exposure to those frauds.

The Planning Phase

Step 3: Build a Profile of Potential Frauds

Fraud examiners might consider referring to the ACFE's occupational fraud and abuse classification system, otherwise known as the fraud tree, when building a profile of potential frauds.



ATTACHMENT

A copy of the fraud tree is provided as an attachment to this course.

The Planning Phase

Step 3: Build a Profile of Potential Frauds

Many organizations, especially larger companies, already have a formal fraud risk assessment process in place. If such is the case, the fraud examiner can simply refer to the outcomes of that process and form a fraud detection approach based on the risks identified. However, if the organization being investigated lacks such a process, the task of conducting a fraud risk assessment from scratch is very time-consuming. It requires the fraud examiner to gain a sufficient understanding of the company's operations so that he can consider the full spectrum of possible fraud scenarios.

The Planning Phase

Step 4: Determine Whether Predication Exists

Data analytics is instrumental in helping a fraud examiner determine predication. *Predication* is the totality of conditions that would lead a rational, professionally trained, and sensible individual to believe a fraud has occurred, is occurring, or will occur. It is the basis upon which a fraud examiner begins an investigation. Internal fraud investigations should not be conducted without proper predication.

The Planning Phase

Step 4: Determine Whether Predication Exists

Suppose someone has submitted a tip that gives the fraud examiner reason to believe a particular fraud has occurred. Before accusing someone of wrongdoing, the fraud examiner must perform some preliminary work, such as data analysis, to determine predication. However, he must do this carefully to avoid defamation.

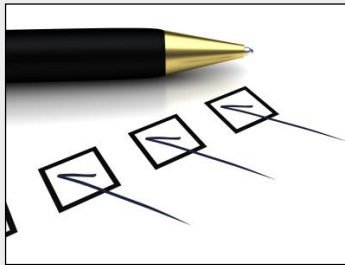
[Defamation](#) is any intentional false communication, either written or spoken, that harms an organization's reputation.

The Planning Phase

Step 4: Determine Whether Predication Exists

To illustrate, imagine a bull's-eye representing a particular fraud. A fraud examiner must stay on the outside of the bull's-eye during the predication check, which means he should avoid conducting any interviews or making accusatory statements before predication is determined. Instead, he should conduct data analysis that does not infringe upon anyone's rights. Because the investigation of fraud deals with the individual rights of others, an investigation must be conducted only with adequate cause or predication.

Lesson 6: The Preparation Phase



Introduction



“Garbage in, garbage out” refers to the philosophy that no matter how sophisticated your data analysis techniques might be, your results are only as valuable as the data you use. It is important to take the time to ensure the quality of your data before performing any tests.

The Preparation Phase

The results of a data analysis test are only as good as the data used for the analysis. Thus, before running tests on data, the fraud examiner must make sure data being analyzed are relevant and reliable for the objective of the engagement. He might be combining the data from different databases or downloading it into a different software that changes its format. The phrase “garbage in, garbage out” is applicable in the preparation phase.

The Preparation Phase

The preparation phase of a data analysis engagement includes the following steps:

- Step 1: Identify the relevant data.
- Step 2: Obtain the data.
- Step 3: Verify the data.
- Step 4: Cleanse and normalize the data.



The Preparation Phase

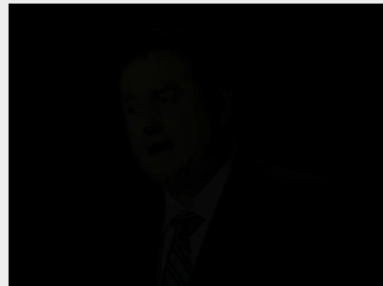
Step 1: Identify the Relevant Data

The first step of the preparation phase of a data analysis engagement is identifying the relevant data. Using the profile of potential frauds as a guide, identify the target data for analysis. Specifically, for each specific fraud scenario assessed to be a high risk to the organization, the fraud examiner should determine which data fields and records would be affected by such a scheme. He should endeavor to obtain as much of this data as is possible, given his time or financial constraints.

The Preparation Phase

Step 1: Identify the Relevant Data

Data analytics is not limited to running a few tests on the general ledger detail. Companies house all types of data that can be useful to a fraud examiner. Listen as Vincent Walden discusses which types of data should be collected.



The Preparation Phase

Step 1: Identify the Relevant Data

When identifying the relevant data to use in the analysis, fraud examiners should determine the logistics involved with obtaining the data, including:

- What fields of the data are available
- Who generates and maintains the data
- Where the data is stored (e.g., electronically, at a remote facility)
- How to receive and store the data
- Control totals needed for verification

The Preparation Phase

Step 2: Obtain the Data

Once the fraud examiner has determined what data are needed, he must prepare and submit a formal request for the desired data, outlining the specifics determined in the previous step. Depending on the objectives of the engagement and the operations of the organization, he might be given a file containing the data to work with, or he might be provided read-only access to the data within the organization's information system.

The Preparation Phase

Step 2: Obtain the Data

As previously mentioned, obtaining the data is one of the most challenging parts of the data analysis process. In some cases, obtaining the data involves working with overloaded or uncooperative data managers, IT departments, or other parties. Other potential obstacles at this stage include: data housed in different systems or on different platforms, and manually maintained data.

The Preparation Phase



Step 3: Verify the Data

The next step of the preparation phase is to verify that the data obtained is complete and legitimate. First, the fraud examiner should ensure that the data analysis software to be used is able to open and read the data as provided. Then, he should validate that the data received contains all requested fields and records.

The Preparation Phase

Step 3: Verify the Data

The fraud examiner should assume he has successfully transferred the desired data onto his system and that the software he plans to use is functioning correctly. Now he must perform some tests to verify the integrity of the data, including:

- Confirming control totals
- Confirming the period covered by the data is appropriate
- Sorting the file in ascending or descending order to test for leading or lagging errors

The Preparation Phase

Step 3: Verify the Data

- Confirm the format of data in format-specific fields, such as date fields and numeric fields.
- Check for blank fields where information should be entered and obtain an explanation for why this information might be missing.
- Check for inappropriate duplicate fields or records.
- Check for gaps in applicable fields, including system-assigned identifying record numbers or check numbers, to identify missing records.

The Preparation Phase

Step 4: Cleanse and Normalize the Data

Listen as Brenda Buetow describes the process of scrubbing a set of data in preparation for analysis.



The Preparation Phase



Step 4: Cleanse and Normalize the Data

Fraud examiners might need to cleanse and convert the data to a format suitable for analysis before executing any tests. For example, certain field formats (e.g., date, time, currency, etc.) might need to be modified to make the information consistent and ready for testing.

The Preparation Phase

Step 4: Cleanse and Normalize the Data

The data must also be normalized so that all data being imported for analysis can be analyzed consistently. Common data fields from multiple systems must be identified, and data must be standardized. In normalizing the data for analysis, table layout, fields/records, data length, data format, and table relationships are all important to consider.

The Preparation Phase

Step 4: Cleanse and Normalize the Data

It is not uncommon for data formats to change when transferred from one program to another. For example, if the fraud examiner has a column with vendor ID numbers, he should ensure that all numbers are consistent.

The Preparation Phase

Step 4: Cleanse and Normalize the Data

Additionally, the following inconsistencies in the data should be considered:

- Known errors
- Blanks or missing data
- Duplicated data
- Special/unreadable characters in the data
- Other unusable entries

Lesson 7: The Testing and Interpretation Phase



Introduction

The third phase of the data analysis process is the testing and interpretation phase. If the planning and preparation stages have been conducted effectively, the testing and interpretation phase should yield results that are helpful in uncovering red flags of fraud in the data.

The Testing and Interpretation Phase

In the testing and interpretation phase, fraud examiners perform the appropriate data analysis techniques and look for anomalies or trends in the data. In order to do this effectively, data must first be *organized* so it can be *analyzed*.



The Testing and Interpretation Phase

Organizing the Data

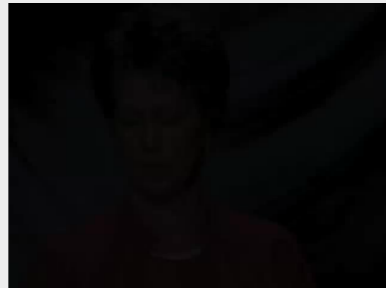
Data should be organized by fraud examiners in such a manner that it can uncover patterns that are consistent with specific fraud schemes identified during the planning phase. Grouping data into similar groups can make it easier for the fraud examiner to spot outliers. Examples of data groups might include:

- Location
- Business unit
- Time period
- Dollar value

The Testing and Interpretation Phase

Organizing the Data

Kelly J. Todd, CFE, CPA is a director of forensic services and specializes in the area of forensic accounting, fraud examination, and litigation support. Listen to this video as she explains what a fraud examiner is looking for during the testing and interpretation phase of the data analysis process.



The Testing and Interpretation Phase

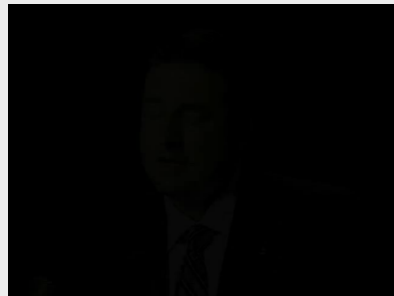
Analyzing the Data

When using data analysis to detect fraud, fraud examiners should strive to organize and analyze the data in a way designed to uncover patterns that are consistent with the specific fraud scenarios identified during the planning stage. The specific data analysis techniques that the fraud examiner should perform given the fraud risks he has identified will be discussed later in this course. For now, we will discuss analyzing data from a high-level perspective.

The Testing and Interpretation Phase

False Positives

Another way to avoid generating false positives in testing is to employ more sophisticated forensic analytics tests. In this video, Vincent Walden talks about the importance of going beyond traditional rules-based tests.



The Testing and Interpretation Phase

At this point in the data analysis process, potential frauds have been identified; the relevant data have been identified, obtained, and verified; and the fraud examiner has performed tests on the data and interpreted the test results. The final phase of the data analysis process is the post-analysis phase.

Lesson 8: The Post-Analysis Phase



Introduction

The data analysis tests run in the previous phase can reveal many potential areas of exploration. After all, the ultimate goal of data analysis is to *narrow the scope* of a fraud investigation. The tests performed should hopefully guide the fraud examiner toward data that look suspicious or unusual. Once a fraud examiner knows where the anomalies lie, he can look into source documents, bank transactions, vendor communications, and so on to figure out whether fraud has occurred. After executing the desired tests, a fraud examiner must determine how to respond to the findings and how to watch for future anomalies. This occurs in the *post-analysis phase*.

The Post-Analysis Phase

After testing and analyzing data, there will likely be potential areas that need further review. After executing the desired tests, fraud examiners should consider the following:

- How to respond to the findings
- How to watch for future anomalies



The Post-Analysis Phase

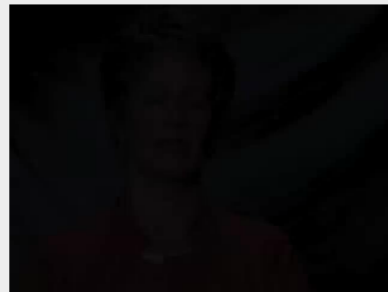
Responding to Findings

The findings of the data analysis engagement are rarely sufficient on their own to conclude that fraud is or is not occurring. Any anomalies identified require follow-up and further examination to determine the reason for the discrepancies in the data. Often, this includes investigatory procedures such as document examination and interviews. The specific resultant actions should be based on the fraud scenarios in question and the particular red flags identified in the data.

The Post-Analysis Phase

Responding to Findings

Listen as Kelly Todd describes a time that data analytics revealed some very unusual invoice activity that turned out to have a reasonable explanation.



The Post-Analysis Phase

Responding to Findings

When responding to analysis findings, exercise caution. Red flags do not always indicate an actual instance of fraud. The point of a data analysis engagement is to identify anomalies and conduct further investigation of these anomalies. This might require obtaining physical documents, conducting interviews with employees and vendors, or examining bank statements. As in the example just described by Kelly Todd, these anomalies might turn out to be nothing. However, fraud examiners need to eliminate the possibility that the anomalies point to fraud.

The Post-Analysis Phase

Responding to Findings

The terms and objectives of the engagement dictate whether a formal report should be issued and, if so, the form and audience of the resulting report. A report should be easy to understand, summarize the analysis findings, and provide insightful information for the readers.

The Post-Analysis Phase

Monitoring the Data

Based on the results of the analysis and the needs of the organization, the fraud examiner should work with management to determine whether any of the tests performed should be repeated on a periodic or continuous basis. In areas deemed to be at heightened risk for fraud, continuous monitoring can provide increased assurance of early detection of any fraud schemes, and can even serve as a deterrent to misconduct. Recall that the perception of detection is the greatest deterrent to fraud.

ACFE ONLINE SELF-STUDY CPE COURSE

Part III: Data Analysis Techniques



Introduction

Data analysis techniques can be implemented in an infinite number of ways. You have to take into consideration what fraud risks are present, what data is available, and what predication exists before you can determine which tests you want to run. Part III of this course is designed to teach you basic data analysis techniques that you can apply to almost any investigation. Part III also includes data analytic techniques that can assist you with identifying specific fraud schemes. Additional techniques that will be discussed include ratio analysis, textual analytics, and other data analysis techniques such as Benford's Law.

Lesson 9: Basic Analytical Procedures



Introduction

There are a number of basic tests that fraud examiners and auditors frequently perform that are successful at detecting anomalies in a data set. This lesson will list some of these tests and discuss how they are useful in detecting fraud. It is important to remember, however, that the use of a particular technique can vary depending on the type of fraud being investigated. Sometimes a combination of these techniques might be required to detect fraudulent activity.

Basic Data Analysis Techniques

Data analysis involves the use of specific techniques to extract and examine data electronically. Numerous techniques are available, each providing a means to identify anomalies that might indicate fraud.



Basic Data Analysis Techniques

Common data analysis techniques include the following:

- Applying filters
- Checking for gaps
- Looking for duplicates
- Sorting the data
- Joining and matching
- Round-dollar payments
- Identifying amounts below a threshold
- Identifying unusual times and dates

Basic Data Analysis Techniques

Applying Filters

Depending on the size of the organization and the time period under investigation, a data set might include thousands or even millions of records. Fraud examiners, depending on the case, probably do not need all of this data. A *filter* identifies only those records meeting user-defined criteria. After obtaining an overview of the data, a fraud examiner can drill down into the details by specifying criteria to isolate certain records for review. The use of display criteria can focus attention on transactions outside of the ordinary and reduce review (and processing) time. The criteria can be used singly or in combination.

Basic Data Analysis Techniques

Example

While the actual application of filters is easy, properly defining the criteria before applying the filter can be difficult. In this example, the fraud examiner is only interested in looking at checks written, but he must isolate that data from the entire disbursements general ledger detail. He filtered the “Check No” column to remove the ACH payments, foreign drafts, and wire transfers from review.

The screenshot shows a filter dialog box for the 'Check No' column. The 'Number Filters' section is expanded, showing a list of checkboxes for different payment types. The following table represents the data shown in the dialog:

Check No	Vendor ID	Vendor
Sort Smallest to Largest		Chase Bank
Sort Largest to Smallest		William Ric
Sort by Color		MCI Group
Clear Filter From "Check No"		HLB Proxy,
Filter by Color		Rebecca O'
Number Filters		ADP, Inc.
☒ 41166		Austex Prin
☒ 41167		Central Tex
☒ 41168		iStockphoto
☒ 41169		Wall Street
☒ 41170		AAC LLP
☒ 41171		MailFinanc
☒ 41172		Ozarka Spr
☐ ACH PAYMENT		Stella Inc.
☐ FOREIGN DRAFT		Mainfreigh
☐ WIRE TRANSFER		Hart & Turn
		Weirich and

Buttons: OK, Cancel

Basic Data Analysis Techniques

Checking for Gaps

Another important test is the search for missing items in a series or sequence of consecutive numbers. Although the completeness of the data is important for any fraud analysis, it can also be an indicator that a fraud has occurred. When searching for fraud, identifying what is not there can often be as important as identifying what is there. A fraud examiner should look for expected items to ensure they do exist and investigate further if he realizes there is missing data. Some examples of data sets where this tool is useful include check numbers, purchase order numbers, and inventory tags.

Basic Data Analysis Techniques

Looking for Duplicates

Looking for unexpected duplicates in a data set is another way to uncover red flags. This technique can quickly review the file, or several files joined together, to highlight duplicate values of key fields. In many systems, the key fields should contain only unique values (no duplicate records). Although not necessarily proof of fraud, the presence of duplicate check numbers, direct deposit numbers, invoices, vendor names, and other data often merits investigation.

Basic Data Analysis Techniques

Looking for Duplicates

Duplicate transactions are not necessarily indications of problems with data completeness or fraud. There might be a perfectly valid reason for the duplicate records. As always, a sound understanding of the data is necessary before drawing conclusions. For example, two different payments coded to the same invoice number is a red flag that perhaps one invoice was paid twice. However, a reasonable explanation for this finding would be that the invoice was paid in two partial payments. In this case, finding the original invoice copy and reading the payment terms would determine whether there was wrongdoing.

Basic Data Analysis Techniques

Sorting the Data

Sorting arranges the data in a file in ascending or descending order based on one or more specified key fields. The commands can arrange information on any number of key fields quickly and easily. The person committing the fraud might rely on the sheer number of transactions to hide an inappropriate series of items. However, unusual transactions can be found simply by sorting on a field, such as the date or account number.

Basic Data Analysis Techniques

Sorting the Data

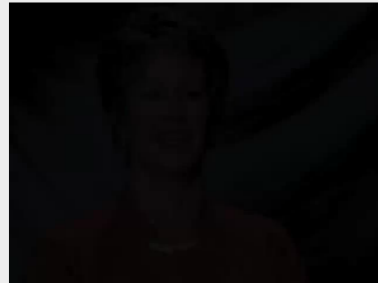
Often, fraud examiners find irregularities in the records at the beginning or end of a sorted file. For example, if the fraud examiner sorts the inventory listing by acquisition date, he might find some really old inventory that does not belong on the books. Other irregularities to look for include:

- Transaction values outside of the normal range for the field
- Character data in numeric fields (or vice versa)
- Records with blank field values
- Payee or vendor names starting with blanks or unusual characters

Basic Data Analysis Techniques

Sorting the Data

In this video, Kelly Todd describes an instance when she discovered check fraud by simply sorting the check register.



Basic Data Analysis Techniques

Joining and Matching

Often, the success of a fraud depends on the separation of two key pieces of information. Therefore, a fraud examiner might want to link together two files or databases and compare them. If the details of a fraud are contained in separate files, the crime might become obvious when the two pieces of information are viewed together. Once the fraud examiner combines two files, he must define a relationship, and compare and contrast the data. You can find an example of this technique in the following lesson on billing schemes.

Basic Data Analysis Techniques

Round-Dollar Payments

Some creative fraudsters go to great lengths to conceal their schemes and make fraudulent activity appear legitimate. However, this is often not the case. While most real-world cash transactions do not occur in simple round numbers, fraudsters often use such figures for simplicity. An unusual amount or regular occurrence of round-dollar payments is a red flag in many fraud schemes.

Basic Data Analysis Techniques

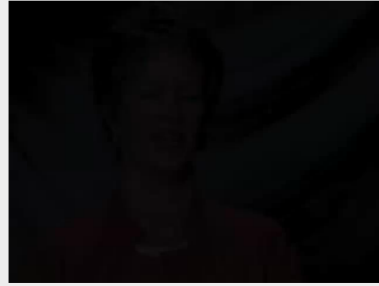
Identifying Amounts Below a Threshold

Most organizations have policies that disbursement amounts above a certain threshold require special consideration. For example, checks greater than \$5,000 might require two signatures, or invoices greater than \$1,000 might need approval from a senior manager before they can be submitted for payment. Fraudsters who are aware of these thresholds often tailor their schemes to fall just below the approval limit. In one corruption case, several payments of \$9,999 turned out to be bribes. The payment approval threshold at that company was \$10,000, so the payments were not discovered until long after they were made.

Basic Data Analysis Techniques

Identifying Amounts Below a Threshold

Listen as Kelly Todd describes a fraud that was discovered by looking for transactions that fell just under the board approval threshold.



Basic Data Analysis Techniques

Identifying Unusual Dates and Times

Fraud is often committed during non-business hours to avoid detection. It is important to identify any transactions that happen on unusual dates or at strange times. For example, suppose you have an employee who is the first to get to the office in the morning and the last to leave at night. They might come in and work weekends and holidays, and never take vacation. Some might be impressed with this individual, thinking he must be an incredibly hardworking and diligent employee. However, fraud examiners should not be so naïve. They should perform testing to see if this person might have an ulterior motive for working these hours.

Lesson 10: Data Analytics and Scheme-Specific Frauds



Introduction

One of the keys to successfully using data analysis is to highlight the red flags typically associated with a particular fraud scheme. The next few lessons provide examples of scheme-specific frauds and their red flags. A list of data analysis tests tailored to each fraud scheme category is also provided. Please note, however, that the lists of red flags and data analysis tests are not intended to be the *only* red flags and data analysis tests for these specific schemes. They should be used as a guide so that they can be applied to organizations based on their own makeup.

Occupational Fraud Schemes

As mentioned earlier, there are three main categories for occupational fraud schemes. Under each of these categories are subcategories that refer to a specific type of scheme (e.g., cash, bribery, inventory). The next few lessons discuss fraud-specific schemes that relate to some of the schemes listed in the fraud tree. Please note, however, that not all schemes listed in the fraud tree will be discussed in this course.

Lesson 11: Billing Schemes



Introduction

Billing schemes attack the victim company's cash disbursements and payables cycle, causing the company to pay for goods or services that are nonexistent, overpriced, or unnecessary. The purpose of most billing schemes is to generate cash for the fraudster. In a typical scheme, the fraudster creates false support for a fraudulent purchase. The false support documents, which can include invoices, purchase orders, purchase requisitions, receiving reports, and so on, cause the victim organization to issue a payment. The fraudster then collects the payment, thereby reaping an illegal gain.

Billing Schemes

In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, billing schemes made up nearly 22 percent of the fraud cases reported and resulted in a median loss of \$100,000. Common billing schemes include:

- Invoicing via a shell company
- Invoicing via an existing vendor
- Personal purchases with company funds

Billing Schemes



Invoicing Via a Shell Company

A *shell company* is a business entity that typically has no physical presence, no employees, and generates little, if any, independent economic value. Although shell companies are often created for legitimate purposes, sometimes they are formed for fraudulent ones.

Billing Schemes

Invoicing Via a Shell Company

When created for fraudulent purposes, shell companies might be just an entity with a fraudulent name and PO Box in which the fraudster uses to collect monies from false billings. Once the shell company is formed, the fraudster creates fictitious invoices in the name of the shell company and sends them to his employer for payment.

Billing Schemes

Invoicing Via an Existing Vendor

Some billing schemes involve legitimate vendors rather than shell companies. In these cases, the fraudster creates false billings on behalf of a vendor with whom the company has done business in the past. Thus, several red flags that often show up in shell company schemes (e.g., questionable addresses or first-time vendors) do not necessarily appear when the false billings seem to be from a legitimate company. The major methods of billing schemes involving existing vendors are:

- False invoicing for non-accomplice vendors
- Pay-and-return schemes

Billing Schemes

Invoicing Via an Existing Vendor

In a *false invoicing for non-accomplice vendor scheme*, the fraudster either creates a fake invoice for a vendor who provides a lot of services or goods to the victim organization or resubmits an invoice that has already been paid. The fraudster then intercepts the resulting payment before it reaches the non-accomplice vendor. The victim is the fraudster's employer, which pays for goods or services that it does not receive.

Billing Schemes

Invoicing Via an Existing Vendor

In a *pay-and-return scheme*, the fraudster purposely overpays a legitimate invoice to an existing vendor. Once the overpayment has been made, the fraudster contacts the vendor and explains that there was a mistake. The fraudster then asks the vendor to return the excess payment amount to his attention so the problem can be resolved. When the vendor returns the overage, the fraudster pockets it.

Billing Schemes

Personal Purchases with Company Funds

Rather than engage in false billings, some fraudsters make personal purchases of goods or services and then charge them to their employer as though the purchases were business expenses. This type of scheme is often accomplished in one of two ways. One method is to run a voucher for a personal purchase through the payables system. Here, the fraudster claims that the goods or services are being bought on the company's behalf. The second method is to purchase personal goods or services on the company's credit card or on a company credit account.

Billing Schemes

Red Flags



- Vendors with incomplete profiles
- Duplicate payments
- Round-dollar payments
- Purchases without purchase orders
- Invoices directly below approval limits
- Unusual journal entries to inventory or cost of sales accounts
- Excessive purchases of particular items
- Excessive purchases from particular vendors
- Unusually high usage of purchasing cards by certain employees

Billing Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of billing schemes, please see the *Red Flags* attachment.

Tests for Billing Schemes

Tests fraud examiners might consider running when investigating a possible billing scheme include, but are not limited to:

- Review changes to the vendor master file, including new vendors added and address changes.
- Identify vendors with significant increases or decreases in activity.
- Extract vendors with incomplete profiles, especially those with missing telephone numbers or tax ID numbers.

Tests for Billing Schemes

- Review transactions just under authorization thresholds.
- Review payments with little or no sequence between invoice numbers.
- Review transactions posted during non-business hours.

Tests for Billing Schemes



ATTACHMENT

For a more comprehensive list of data analysis tests to run for billing schemes, please see the *Data Analysis Tests* attachment.

Lesson 12: Check Tampering Schemes



Introduction

Check tampering is a type of fraudulent disbursement scheme in which an employee either: (1) prepares a fraudulent check for his own benefit, or (2) intercepts a check intended for a third party and converts the check for his own benefit. Check tampering schemes depend on factors such as access to the company check stock or online payment system; bank statements; and cash disbursements journal, as well as the ability to forge signatures or alter other information on the face of the check.

Check Tampering Schemes

In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, check tampering schemes made up nearly 11 percent of the fraud cases reported and resulted in a median loss of \$158,000. The following are four methods used to commit check tampering:

1. Forged maker schemes
2. Forged endorsement schemes
3. Altered payee schemes
4. Authorized maker schemes

Check Tampering Schemes

Forged Maker Schemes

In this type of scheme, fraudsters obtain a blank check from their employer and make it payable to themselves, an accomplice, or a third party. The signature is then forged based on an authorized signatory.

Check Tampering Schemes

Forged Endorsement Schemes

In this scheme, an employee seizes an outgoing check intended for a third party and alters the check by forging the intended payee's name on the endorsement line of the check. The fraudster might intercept the check after it is signed but before it is delivered, and re-route the check to a location where it can be retrieved. Another method is to steal the check once it has been returned to the company as undeliverable.

Check Tampering Schemes

Altered Payee Schemes

In an altered payee scheme, the fraudster intercepts a signed check and alters the name of the payee on the stolen check. The fraudster might use his own name, a fictitious entity, or some other name.



Check Tampering Schemes

Authorized Maker Schemes

This check tampering scheme occurs when an employee with check-signing authority abuses the trust of his employer by writing fraudulent checks on the organization's accounts. The perpetrator can write and sign fraudulent checks without having to alter the check or forge a signature.

Check Tampering Schemes

Red Flags

- Checks issued out of sequence
- Out-of-range checks
- Duplicate check numbers
- Non-payroll checks made out to an employee
- Unusually high number of checks made out to "Cash"



Check Tampering Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of check tampering schemes, please see the *Red Flags* attachment.

Tests for Check Tampering

Tests that can be used to detect check tampering include, but are not limited to:

- Identify checks with no corresponding invoices.
- Review transactions for round-dollar payments.
- Identify purchases without purchase orders by vendor and issuer.
- Review transactions by user ID for unauthorized or unusual users.

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ATTACHMENT

For a more comprehensive list of data analysis tests to run on check tampering schemes, please see the *Data Analysis Tests* attachment.

Lesson 13: Payroll Schemes



Introduction

A payroll scheme is any scheme in which an employee causes his employer to issue a payment by making false claims for compensation. For example, an employee might claim overtime for hours not worked or add fictitious payees to the payroll register. In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, payroll schemes made up nearly 8 percent of the fraud cases reported and resulted in a median loss of \$90,000.

Payroll Schemes

A payroll scheme differs from a billing scheme in that the fraudster causes his employer to issue him a *compensation* payment to which he is not entitled. There are three main categories of payroll fraud:

- Ghost employee schemes
- Falsified wages
- Commission schemes

Payroll Schemes

Ghost Employee Schemes

A *ghost employee* is someone who is on the payroll register but who does not actually work for the company. The ghost can be a real person—such as a former employee—or a fictitious person. In a ghost employee scheme, the fraudster adds a ghost to the payroll register (or fails to remove the ghost from the register, assuming that the ghost is a former employee), and then issues, collects, and converts (if necessary) a fraudulent payment to the ghost. When the ghost employee is an accomplice of the fraudster, conversion of the fraudulent payment is made easier.

Payroll Schemes

Falsified Wages Scheme

In a falsified wages scheme, the fraudster manipulates compensation-related factors in order to increase his pay. Factors subject to manipulation include gross salary, pay rate, net pay, number of hours worked, or number of parts produced. Generally, there are fewer opportunities for salaried workers to perpetrate these schemes than there are for hourly or piecemeal workers, because the pay of salaried workers is not usually based on variable factors such as hours worked or parts produced.

Payroll Schemes

Commission Schemes

A commission scheme occurs when the fraudster alters his commission rate or falsely reports the amount of sales he made. When falsely reporting his sales, he might inflate the sales price or quantity sold, include sales to fictitious customers, or report sales in improper periods (e.g., to earn a higher commission rate that might be paid during certain periods).

Payroll Schemes

Red Flags

- Multiple employees using the same bank account for direct deposit
- Invalid or blank Social Security numbers (SSNs)
- Unsupported adjustments to gross pay
- Unusually high commission amounts in relation to sales



Payroll Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of payroll schemes, please see the *Red Flags* attachment.

Tests for Payroll Schemes

There are many data analysis techniques that can be employed to uncover payroll fraud schemes. Since the information used to calculate salaries and wages can be found in the employee master file, combining the payroll records with the human resources files is a useful way to identify discrepancies. Additionally, since commissions are based on sales, fraud examiners can compare the payroll records for employees who receive commission to the sales records. You might recall from the previous lesson that the practice of putting together two different data sets and analyzing the combined data is called *joining and matching*.

Tests for Payroll Schemes

Listen as Brenda Buetow describes how joining and matching data can help detect a payroll fraud scheme.



Tests for Payroll Schemes

Tests that can be used to detect payroll schemes include, but are not limited to:

- Compare salaried employees' gross pay from one period to the next.
- Identify changes in salary amounts.
- Compare pay rates in the payroll system to those in the HR files.
- Compare bonus payments to budgeted or prior year amounts.



ATTACHMENT

For a more comprehensive list of data analysis tests to run on payroll schemes, please see the *Data Analysis Tests* attachment.

Lesson 14: Expense Reimbursement Schemes



Introduction

In an expense reimbursement scheme, the fraudster makes an improper claim for the reimbursement of unallowable or fictitious expenses. In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, expense reimbursement schemes made up nearly 14 percent of the fraud cases reported and resulted in a median loss of \$40,000.

Expense Reimbursement Schemes

Expense reimbursement schemes are one of the five categories under the fraudulent disbursements umbrella. The following are the four types of expense reimbursement fraud schemes:

- Mischaracterized expenses
- Fictitious expenses
- Overstated expenses
- Multiple reimbursements

Expense Reimbursement Schemes

Mischaracterized Expenses

In a *mischaracterized expense reimbursement scheme*, an employee submits non-business expenses to his employer for reimbursement. This practice might involve claiming a personal expense or an expense that was legitimately made but is not in compliance with the guidelines of the organization's policy for reimbursement. Examples of mischaracterized expenses include claiming personal travel as a business trip, listing dinner with a friend as business development, and so on.

Expense Reimbursement Schemes

Fictitious Expenses

Sometimes the fraudster creates purely fictitious expenses when there were none. He might do this by creating his own phony receipt or by using someone else's receipt. The fraudster might even go so far as to obtain a stack of blank receipts from a vendor and then use it to manufacture support for his fictitious expenses. These schemes can be fairly easy to get away with if there are weak controls over expense reimbursement or if the fraudster's supervisor does not closely examine his expense reports.

Expense Reimbursement Schemes

Overstated Expenses

Overstated expense reimbursement schemes occur when the fraudster inflates the dollar amount of legitimate business expenses and then submits the fraudulent amounts to his employer for reimbursement. The dollar amount can be inflated through altering an existing receipt, creating a phony receipt, asking a vendor to produce the receipt for a higher amount, or some other means.

Expense Reimbursement Schemes

Multiple Reimbursements

Multiple expense reimbursement schemes occur when the fraudster submits the same business expense for reimbursement over and over again. These schemes are especially easy to commit by employees who frequently travel for work and have similar expenses on every report. In order to minimize the chance of this scheme being detected, the fraudster might space out his expense reimbursement submissions or submit them to different reviewers.

Expense Reimbursement Schemes

Red Flags

- Expenses submitted long after they were incurred
- Employee expenses paid to vendors normally associated with personal purchases
- Expenses that are consistently rounded off
- Expenses that are consistently for the same amount
- Excessive mileage claims



Expense Reimbursement Schemes

Red Flags

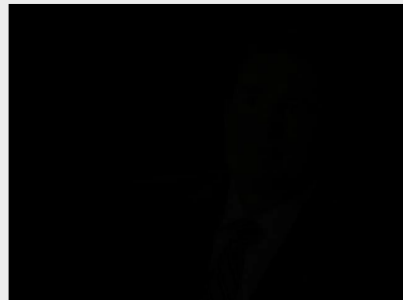


ATTACHMENT

For a more comprehensive list of red flags of expense reimbursement schemes, please see the *Red Flags* attachment.

Tests for Expense Reimbursement Schemes

As discussed, there are several different types of expense reimbursement fraud schemes. Listen as Vincent Walden describes one of the tests he commonly uses to detect travel and expense fraud.



Tests for Expense Reimbursement Schemes

Tests that can be used to detect expense reimbursement schemes include, but are not limited to:

- Summarize expense reimbursements by employee.
- Age employee payments by check date.
- Extract all reimbursements just below approval limit.
- Identify overlapping travel claims.
- Extract all round-dollar reimbursements.

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ATTACHMENT

For a more comprehensive list of data analysis tests to run on expense reimbursement schemes, please see the *Data Analysis Tests* attachment.

Lesson 15: Theft of Cash Receipts Schemes



Introduction

Cash is the most liquid asset and therefore the most vulnerable to misappropriation. In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, cash larceny comprised 8 percent of the reported cases, with a total median loss of \$90,000. This lesson addresses two categories of cash fraud schemes:

- Theft of cash on hand
- Theft of cash receipts

Theft of Cash on Hand

In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, theft of cash on hand comprised nearly 11 percent of the reported cases, with a total median loss of \$25,000. Theft of cash on hand occurs when fraudster takes cash on hand that is kept on the company's premises. Examples of theft of cash on hand include:

- Stealing from a petty cash fund
- Stealing cash from a vault or safe

Theft of Cash Receipts

Theft of cash receipts schemes are what we typically think of as outright stealing cash. The fraudster does not rely on the submission of phony documents or the forging of signatures; he simply grabs the cash and takes it. There are two types of cash receipts schemes:

- Skimming
- Cash larceny

Theft of Cash Receipts

Skimming

In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, skimming comprised nearly 11 percent of the reported cases, with a total median loss of \$53,000. Skimming schemes all follow the same basic pattern.



Theft of Cash Receipts

Example



An employee steals incoming funds before they are recorded in the victim organization's books. Since the missing funds are never recorded, skimming schemes leave no direct audit trail, making them difficult to detect. Skimming schemes usually target cash, but can also target receivables.

Theft of Cash Receipts

Skimming

Skimming receivables is more difficult to conceal than skimming sales. The victim organization will usually notice if expected receivables are not received and recorded—and so are customers who receive invoices for amounts previously paid.

Theft of Cash Receipts

Skimming

Techniques used to conceal skimmed receivables include:

- Lapping
- Altered or destroyed records
- Fraudulent write-offs
- Debits to the wrong account
- Force balancing



Theft of Cash Receipts

Cash Larceny

The other type of cash receipts scheme is cash larceny. *Cash larceny* is the theft of cash that has already been recorded as received on a victim company's books. Cash larceny can take place in any instance in which an employee has access to cash.

Theft of Cash Receipts

Cash Larceny

While the circumstances under which an employee might steal cash are nearly limitless, larceny schemes frequently involve the theft of cash:

- At the point of sale
- From incoming receivables
- From the victim company's bank deposits

Theft of Cash Receipt Schemes

Red Flags

- Unusual entries to cash accounts
- Unusual entries to accounts receivable
- Invoices with partial payments
- Customers with credit balances
- Unusually high uncollectable accounts



Theft of Cash Receipt Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of theft of cash receipt schemes, please see the *Red Flags* attachment.

Tests for Theft of Cash Receipt Schemes

Tests that can be used to detect theft of cash receipt schemes include, but are not limited to:

- Extract customers with credit balances.
- Identify higher-than-average uncollectable accounts.
- Identify accounts with even balances.
- Identify gaps in the sequencing of invoices generated.



ATTACHMENT

For a more comprehensive list of data analysis tests to run on theft of cash receipt schemes, please see the *Data Analysis Tests* attachment.

Lesson 16: Noncash Asset Schemes



Introduction

Noncash assets, or inventory and all other assets, are common targets for theft, although less commonly stolen than cash. In the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, noncash asset misappropriations (including securities, information, and physical assets) accounted for 19 percent of all cases reported and had a median loss of \$70,000.

Noncash Asset Schemes

Noncash asset schemes might involve theft or *larceny* of inventory and all other assets, either for personal use or for sale. Inventory that is relatively small in size, high in value, or readily salable is most vulnerable to fraud. For example, in the restaurant industry, alcohol and meat inventory are common targets of theft.

Noncash asset schemes might also include the *misuse* of assets, such as taking company supplies home for personal use and bringing them back on site.

Misuse of Inventory and All Other Assets

The misuse of noncash assets might not seem to be as severe as outright theft of noncash assets, but over time it can affect the company's bottom line. Using company noncash assets for personal use can cause equipment to depreciate in value more quickly due to wear and tear. For example, an employee of a print shop might use the shop's printing presses without permission to create business cards for customers of his side business. Over time the printing presses might become less efficient due to overuse by the employee for personal reasons.

Theft of Inventory and All Other Assets

Larceny or the stealing of inventory and all other assets include the following:

- Asset requisitions and transfer schemes
- False sales and shipping schemes
- Purchasing and receiving schemes
- Unconcealed larceny schemes



Theft of Inventory and All Other Assets

Asset Requisitions and Transfer Schemes

Some fraudsters falsify internal documents that deal with the requisition or internal movement of inventory so that they can steal it. They use these documents to justify the transfer or re-allocation of inventory that they otherwise might not be able to access. Examples of this type of scheme include overstating the amount of materials required for a project, requesting merchandise for an improper purpose, and fraudulently transferring inventory from one site to another within the victim organization. In a related scheme, some fraudsters scrap good inventory and then sell it later.

Theft of Inventory and All Other Assets

False Sales and Shipping Schemes

Some fraudsters misappropriate inventory by creating false sales orders or other shipping documents. These false documents represent sales to fictitious persons, fictitious companies, or accomplices of the fraudster. The victim organization's shipping department sends out the targeted inventory as though the inventory has been sold; in reality, however, there is no sale. The merchandise is delivered to the fraudster or his accomplice.

Theft of Inventory and All Other Assets

Purchasing and Receiving Schemes

Another relatively common method for stealing inventory is to falsify receiving reports or skim inventory items from incoming deliveries. For example, a warehouse supervisor or clerk might falsify a receiving report to show that a shipment was short or that some of the goods in the shipment were defective. This fraudster would then steal the "missing" or "substandard" merchandise.

Theft of Inventory and All Other Assets

Unconcealed Larceny

The most basic method for stealing inventory is *unconcealed larceny*, in which the fraudster takes inventory from the victim organization's premises without attempting to conceal the theft in the victim's books and records. In other schemes, the fraudster might create false documentation to justify the shipment of merchandise or tamper with inventory records to conceal missing items. In some unconcealed larceny schemes, however, fraudsters are not as discreet. The culprit in these crimes takes inventory without trying to justify its absence.

Noncash Asset Schemes

Red Flags

- Unusually high amount of written-off inventory
- Duplicate inventory items
- Inventory price higher than retail price
- Unusual variances from standard pricing
- A disproportionate increase in cost of goods sold compared to sales



Noncash Asset Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of noncash asset schemes, please see the *Red Flags* attachment.

Tests for Noncash Asset Schemes

Tests that can be used to detect theft or misuse of noncash asset schemes include, but are not limited to:

- Identify negative receipt quantities.
- Calculate stock inventory turnover by item.
- Identify items below the standard margins.



ATTACHMENT

For a more comprehensive list of data analysis tests to run on noncash asset schemes, please see the *Data Analysis Tests* attachment.

Lesson 17: Corruption Schemes



Introduction

Corruption schemes involve the influence that employees use in a way that violates their duty to their employer in order to receive a benefit for themselves or someone else. Such schemes often involve collusion between the internal employee and an outside party, such as a vendor or customer; consequently, they can be extremely difficult to detect. Corruption is a serious concern to any company that operates in multiple countries around the world. In some countries, although it might be illegal, corruption is still a common business practice.

Corruption Schemes

Corruption schemes made up close to 35 percent of occupational frauds researched in the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*. The median loss an organization suffered from a corruption scheme was \$200,000. Corruption schemes include:

- Bribery
- Illegal gratuities
- Economic extortion
- Conflicts of interest

Corruption Schemes

Bribery

In a bribery scheme, the fraudster solicits, offers, promises, gives, or receives money or some other item of value to illegally or unethically influence a business decision or an official act of government. The two major classifications of bribery schemes include:

- Kickbacks
- Bid rigging

Corruption Schemes

Bribery

A *kickback* is an undisclosed payment by a vendor to a corrupt employee of a purchasing company to ensure the company's business. For example, a vendor might submit an inflated invoice and an employee of the victim company makes sure that the inflated invoice gets approved. In return, the vendor kicks back some of the amount overpaid by the victim organization to the employee.

Corruption Schemes

Bribery

Bid rigging occurs when a corrupt employee assists a vendor in winning a contract with the victim organization. The bidding process for vendors can become cutthroat, causing some to seek the assistance of someone on the inside of the organization to gain advantage over the other vendors. This method might help ensure that a vendor wins a sought-after contract, especially if the vendor is willing to pay for this influence with a bribe.

Corruption Schemes

Illegal Gratuities

Illegal gratuities schemes are similar to bribery schemes, except that the fraudster carries them out to reward a decision rather than to influence one. In an illegal gratuities scheme, a decision is made that happens to benefit a specific person or company. This decision is not influenced by any sort of payment. The party who benefits from the decision then rewards the person who made the decision. For example, consider a situation in which an employee awards a construction contract to a specific vendor and then later receives an all-expenses paid vacation from that vendor as a reward. The vacation would be considered an illegal gratuity.

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Corruption Schemes

Economic Extortion

Economic extortion schemes are the “pay up or else” corruption schemes. The fraudster commits economic extortion when he demands payment from another, threatening harm in the event of refusal to pay. For instance, consider a situation in which an employee demands payment from a supplier in order to award a contract; if the supplier refuses, the employee will award the contract to a rival supplier. The employee in this scenario is undertaking an economic extortion scheme.

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Corruption Schemes

Conflict of Interests

A conflict of interest occurs when an employee has an undisclosed economic or personal interest in a transaction that adversely affects his employer. One characteristic of a conflict of interest scheme is the exertion of the employee's influence to the detriment of his organization. Conflict of interests can occur in *purchasing schemes* or *sales schemes*. For example, a corrupt employee might cause his employer to buy something at an inflated price from a company in which he has a hidden interest or cause his employer sell something at an artificially low price to such a company.

Corruption Schemes

Red Flags

- Unusual bonuses or extraordinary payments
- Unorthodox or substantial upfront payments
- A high incidence of invoice payments just below an approval limit
- Variances between contract price and invoice price
- Losing bidders hired as subcontractors



Corruption Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of corruption schemes, please see the *Red Flags* attachment.

Tests for Corruption Schemes

Tests that can be used to detect corruption schemes include, but are not limited to:

- Identify duplicate payments.
- Identify payments with little or no sequence between invoice numbers.
- Compare order quantity to optimal reorder quantity.



ATTACHMENT

For a more comprehensive list of data analysis tests to run on corruption schemes, please see the *Data Analysis Tests* attachment.

Lesson 18: Financial Statement Fraud Schemes



Introduction

Financial reporting schemes occur when the fraudster intentionally misstates or omits financial statement amounts or disclosures to deceive users of the financial statements. According to the ACFE's 2016 *Report to the Nations on Occupational Fraud and Abuse*, financial statement fraud is both the least common and the most costly form of occupational fraud. Financial statement fraud schemes accounted for less than 10 percent of all frauds reported and incurred a median loss of \$975,000. Financial statement fraud typically involves *overstating assets or revenues*.

Financial Statement Fraud Schemes

Financial statement fraud is different from other types of fraud in that the motive is not usually personal gain, but rather to give senior management, the board, or external parties the impression that the company is doing better than it actually is. Financial statement fraud schemes include overstating revenues and assets. These overstatements include the following:

- Timing differences
- Fictitious revenues
- Concealed liabilities and expenses
- Improper asset valuations
- Improper disclosures

Financial Statement Fraud Schemes

Timing Differences and Fictitious Revenues

Under generally accepted accounting principles (GAAP), revenue is recognized when it is realized or realizable and earned. *Timing differences* (involving prematurely recognizing revenue) or *fictitious revenues* are common ways the fraudster can overstate revenue. For example, a fraudster might record a sale for a product that has been ordered, but is not able to be shipped to the customer for another month. According to GAAP, this sale should not be recorded until the ownership transfers from the seller to the buyer. These types of misstatements often occur at the end of a period, when managers are under increased pressure to meet financial targets.

Financial Statement Fraud Schemes

Concealed Liabilities and Expenses

A person committing financial statement fraud might be tempted to manipulate expenses to increase the bottom line. Generally, companies are required to record expenses on the income statement in the period in which the related revenue is earned. This concept is known under GAAP as the *matching principle*. A fraudster might improperly capitalize expenses or just not record the expenses at all. Both of these methods would violate the matching principle and artificially inflate net income.

Financial Statement Fraud Schemes

Concealed Liabilities and Expenses

When a liability is understated, it is often because a corresponding expense was not recorded. Contingent liabilities, such as product warranty obligations or lawsuit settlements, are a popular target for understatement by the fraudster, since these amounts are often based on estimates. Unearned revenue, such as gift card sales, is another attractive area for the fraudster who wants to understate liabilities. Understated liabilities give the impression of a stronger balance sheet than actually exists.

Financial Statement Fraud Schemes

Improper Asset Valuation

The fraudster can overstate assets through a variety of means, including by recording fictitious assets (e.g., accounts receivable); overstating the value of inventory, fixed assets, or intangible assets; or understating allowances for receivables. For example, the fraudster might fail to write off obsolete inventory, or he might capitalize a repair on an asset instead of expensing it, in an effort to inflate the asset balance.

Financial Statement Fraud Schemes

Improper Disclosures

The full disclosure principle under GAAP states that a company's management must disclose all information that would affect a reader's understanding of the financial statements and must not disclose information that would be misleading to users. Disclosures are usually included in the footnotes of financial statements. Fraudsters can manipulate disclosures by failing to include required disclosures, misstating disclosures, or purposefully writing disclosures in a way that is difficult for users to understand.

Financial Statement Fraud Schemes

Red Flags



- Rapid growth or unusual profitability
- Recurring negative cash flows from operations while reporting a growth in earnings
- Significant transactions close to the end of the period
- A shrinking percentage of allowances for sales returns and warranties

Financial Statement Fraud Schemes

Red Flags



ATTACHMENT

For a more comprehensive list of red flags of financial statement fraud schemes, please see the *Red Flags* attachment.

Tests for Financial Statement Fraud Schemes

Tests that can be used to detect financial statement fraud schemes include:

- Review transactions posted at unusual times.
- Prepare account reconciliations.
- Review manual journal entries to accounts that are typically posted automatically.
- Review journal entries by employee to ensure the proper people are posting to accounts they manage.

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ATTACHMENT

For a more comprehensive list of data analysis tests to run on financial statement fraud schemes, please see the *Data Analysis Tests* attachment.

Lesson 19: Financial Statement Analysis



Introduction

Ratios are used to evaluate relationships between two or more measures. When compared over time, to other entities, or to industry averages, ratio analysis provides great insight into an entity's operations. Financial ratios compare two pieces of financial data, such as the value of current assets over current liabilities. However, some ratios are mixed, which means they include both financial and non-financial data.

Financial Statement Analysis

When analyzing financial statements, fraud examiners might compare the relationship between two line items, or the changes in account balances over time. This type of analysis might uncover red flags that point them in the direction of possible fraud. Financial statement analysis includes the following:

- Horizontal analysis
- Vertical analysis
- Ratio analysis

Financial Statement Analysis

Horizontal Analysis

Horizontal analysis is a technique for analyzing the percentage change in individual financial statement items from one period to the next. Dramatic changes from year to year that do not have a clear explanation are red flags. It is important to note that horizontal or vertical analysis will likely not detect small, immaterial frauds.

Financial Statement Analysis

Horizontal Analysis

When conducting horizontal analysis, consider the dollar amount of change, as well as the percentage change. A 5 percent change in an account with a very large dollar amount might actually be much more significant than a 50 percent change in a small account. A fraud examiner should be aware of certain accounts that almost always move in tandem, such as sales and cost of goods sold.

Financial Statement Analysis

Vertical Analysis

Vertical analysis is a technique for analyzing the relationships between the items on an income statement, balance sheet, or statement of cash flows by expressing accounts as percentages. This allows a financial analyst to quickly determine which accounts are the biggest drivers of company performance. Financial statements analyzed in this way are often called *common-size statements* because the dollar amounts are expressed in relation to one another as percentages, and this enhances comparability between other entities.

Financial Statement Analysis

Ratio Analysis

As previously stated, ratio analysis is a means of measuring the relationship between two different amounts. If the financial ratios present a significant change from one period to the next, further examination might be necessary. Even though ratios alone do not prove fraud, they can help fraud examiners detect anomalies when a calculated ratio differs from the value expected based on historical trends.

Financial Statement Analysis

Ratio Analysis

There are many ratios that can be used in a fraud examination to identify red flags. The following slides discuss the ratios listed below:

- Current ratio
- Accounts receivable turnover
- Inventory turnover
- Debt-to-equity



Financial Statement Analysis

Current Ratio

$$\frac{\text{Current Assets}}{\text{Current Liabilities}}$$

This ratio is probably the most commonly used ratio in financial statement analysis. It measures a company's ability to meet present obligations using its liquid (current) assets. In detecting fraud, this ratio can be a prime indicator of manipulation of the accounts involved. For example, embezzlement causes the ratio to decrease. Understated liabilities result in a more favorable ratio.

Financial Statement Analysis

Accounts Receivable Turnover

$$\frac{\text{Net Sales}}{\text{Average Net Receivables}}$$

The accounts receivable turnover ratio measures the number of times accounts receivable is turned over during a period. Suppose a company is reporting fictitious sales. Since the sales never happened, the accounts receivable associated with those sales will sit on the books until they are written off. A decrease in the accounts receivable turnover ratio could be a red flag to a fraud investigator since it is a potential indicator of fictitious receivables.

Financial Statement Analysis

Inventory Turnover

$$\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

This ratio measures the number of times inventory is sold during a period. It is used by analysts to determine purchasing, production, and sales efficiency. Generally, a higher inventory turnover ratio is considered to be favorable. However, this might also indicate fraud.

Financial Statement Analysis

Debt-to-Equity

$$\frac{\text{Total Liabilities}}{\text{Total Equity}}$$

The debt-to-equity ratio is commonly considered by lending institutions. In fact, some debt covenants require that the lender maintain a certain debt-to-equity ratio. The amount of risk associated with an organization is directly related to its debt-to-equity ratio; the lower the ratio, the more favorable it looks to creditors and owners. A fraudster might be tempted to understate liabilities, resulting in a lower debt-to-equity ratio, in order to maintain the requirements of a loan covenant.

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Lesson 20: Benford's Law



Introduction

In the 1920s, physicist Frank Benford made a remarkable mathematical discovery: In a population of multi-digit numbers, the first digit of the numbers is distributed in a predictable way. While you might assume the first digit of any number found in nature would have just as much chance of being 1 as it would being 9, that is not the case. In reality, the first digit of a multi-digit number is 1 approximately 30 percent of the time. Conversely, the first digit of a multi-digit number is 9 only 4.6 percent of the time.

Benford's Law

Interestingly enough, the probabilities decline as the digit increases in value. In other words, 1 is more likely to appear as the first digit in a number than is 2. The pattern continues as the digits increase: 2 is more likely to appear as a first digit than 3; 3 is more likely to appear as a first digit than 4; and so on. Note that 0 is inadmissible as a first digit; therefore, there are nine possible first digits.

Predictable patterns also occur in the second and third digits of multi-digit numbers.

Benford's Law

Digit	Position in Number	
	1 st Digit	2 nd Digit
0	—	11.97%
1	30.10%	11.39%
2	17.61%	10.88%
3	12.49%	10.43%
4	9.69%	10.03%
5	7.92%	9.67%
6	6.70%	9.34%
7	5.80%	9.04%
8	5.12%	8.76%
9	4.58%	8.50%

Digit Distribution

Benford's data suggest that the digit 1 appears as the first digit in a multi-digit number 30.1 percent of the time; the digit 2 appears 17.6 percent of the time; and so on. The digit distributions for the first two digits of naturally occurring four-digit numbers are displayed in this chart.

Benford's Law

Expected Patterns

The law provides the expected patterns of digit frequencies, which should apply to invoice amounts, sales figures, and other sets of naturally occurring numbers. If an auditor finds that the data does not conform to Benford's pattern, further analysis might be needed. The discrepancy could be the result of fraud, errors, some type of inefficiency, or another unusual occurrence. The following example illustrates how Benford's law can be applied to a data set in a fraud investigation.

Benford's Law

Identifying Fabricated Numbers

Many fraudsters fail to consider the counterintuitive pattern of naturally occurring digits when creating false documentation or transactions to cover their tracks. Consequently, conducting a Benford's law analysis on a data set can help identify included numbers that are not legitimate.

Benford's Law

Limitations

Benford's law is not an effective tool for all data sets. The smaller the data set, the less accurate the results become, because there are not enough items in the sample and the rules of randomness do not apply. Additionally, if the data have built-in minimums and maximums, or consist of assigned numbers (such as check numbers), they might not conform to the law's predictions.

Lesson 21: Textual Analytics



Introduction

Traditional data analysis involves running rule-based queries on structured data, such as that contained in transactional databases or financial accounting systems. This type of analysis can yield valuable insight into potential frauds. However, a more complete analysis requires that fraud examiners also consider unstructured data.

Since unstructured data is text-based data, such as that found in emails, documents, reports, presentations, websites, and free text fields in databases (e.g., journal entry descriptions), it makes sense to use *textual analysis* as a means to analyze this data.

What Is Textual Analysis?

Textual analysis is the process used to collect data to explain, understand, or interpret a situation or a person's actions or thoughts. It involves extracting usable information from unstructured text data that can then be used to reveal patterns, sentiments, and relationships that might be indicative of fraud.

Text Mining

Text mining is a method of analyzing text using software to extract usable information from unstructured text data. Through the application of linguistic technologies and statistical techniques—including weighted fraud indicators (e.g., fraud keywords) and scoring algorithms—text analytics software can categorize data to reveal patterns, sentiments, and relationships indicative of fraud. For example, an analysis of email communications might help fraud examiners to gauge the pressures/incentives, opportunities, and rationalizations to commit fraud that exist in an organization.

Text Mining

Textual analytics is an emerging data analysis tool. Listen as Vincent Walden explains what text mining is and how it is useful for fraud examiners.



Textual Analysis Techniques

Textual analysis embraces several different approaches. It includes techniques that match words or phrases; tag and categorize documents; and extract meaningful words or phrases. Fraud examiners can take advantage of the following to perform textual analysis techniques:

- Fraud triangle components
- Keywords
- Sentiment
- Emails
- Social media
- Financial statement text

Textual Analysis Techniques

Fraud Triangle Components

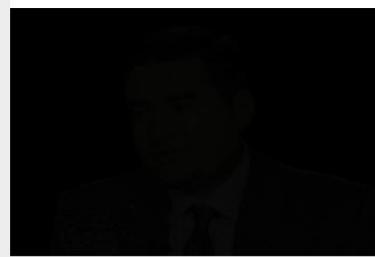
Considering how the three components of the fraud triangle—incentive/pressure, opportunity, and rationalization—can contribute to fraud occurrence is key to understanding how textual analysis can be used to detect and prevent fraud.



Textual Analysis Techniques

Fraud Triangle Components

Listen as Vincent Walden, CFE, CPA, explains how fraud triangle components can be used with textual analysis techniques to uncover fraud schemes.



Textual Analysis Techniques

Keywords

Keywords are the most significant words or phrases in a document. They are the foundation for providing a concise representation of a document's content. In other words, keywords can tell you what a document is about. When used to detect fraud, a targeted analysis using keywords can save a company millions of dollars by raising red flags and suspicious activity before a major fraud is even perpetrated. However, before developing a list of keywords, fraud examiners should be aware of the process involved in the development and management of a keyword list.

Textual Analysis Techniques

Keywords

The fraud triangle is being used in textual analytics to help fraud examiners determine where and why fraud might exist through the use of *keywords*. Pre-defined keyword lists are being used by fraud examiners to detect fraud in areas previously overlooked. This list of keywords might be developed based on known instances of fraud, fraud triangle components, or a combination of both.

Textual Analysis Techniques

Keywords

A keyword search can be incredibly useful when a fraud examiner has a large amount of text and he has identified *fraud buzzwords* that might indicate unscrupulous activity. Take a listing of journal entries, for example. Suppose you are an auditor trying to find an unusual and potential fraudulent journal entry. Examples of keywords to search for in the explanation text might include: *adjust/adjustment, reversal/reverse, catch up, write up, write down, reduce, cut off, rounding, bump, reserve, and early.*

Textual Analysis Techniques

Sentiment Analysis

Sentiment analysis, or *opinion mining*, involves building a system to collect documents and other data to examine opinions. It is focused on identifying “what people think” or “how they feel” about something. This type of analysis can assist management in its decision making or assist fraud examiners in uncovering a fraud scheme. It can serve as a trending tool to look at emerging areas of concern or interest that might indicate suspicious or fraudulent behavior. Sentiment analysis is a text analysis technique that usually involves mining data in emails or social media sites.

Textual Analysis Techniques

Emails

Employees send a massive amount of emails in a given day, many times not considering the content or the context, even if they are the perpetrator or accomplice to a fraud scheme. By reviewing email communications, a fraud examiner can usually gain enough understanding of the context behind the motivation for a fraud scheme, including actions taken, rationale, or intent for making certain decisions.

Textual Analysis Techniques

Emails

When investigating fraud schemes, emails usually provide insight about the timing and intent leading up to a fraud and are one of the first data sources fraud examiners request during investigations. For example, when investigating a fraud related to financial statements, data from emails—not the ledgers—might be more effective in determining when the fraud scheme initially began, how it progressed, and who was involved.

Textual Analysis Techniques

Social Media

Fraud examiners looking for an additional means to detect fraud can take advantage of social media forums. Finding, sorting, and analyzing data from these forums offers them the ability to identify patterns in customer sentiment and trend behaviors as a means to assist in the detection of fraud. While this data might seem overwhelming or even irrelevant to some, social media is the medium where enormous numbers of people come to share their opinions and needs. It is meaningful because it is collected in an environment where people can say whatever they want, whenever and wherever they want.

Textual Analysis Techniques

Social Media

Fraud examiners should be aware that simply gathering snippets of phrases or words from social media outlets is not enough. Human effort is needed to sort and analyze this data into a manageable format so that conclusions can be made based on sentiment. In order to gather this information, however, fraud examiners should be aware of common social media sites and the types of applications dedicated to them. Once this data is gathered, they can perform keyword searches and analyze the results to determine sentiment.

Textual Analysis Techniques

Financial Statement Text

The readability of public company financial statements is a controversial topic, and relevant for fraud examiners. The notes to the financial statements are of particular interest since the language is often difficult to decipher due to the complexity of the underlying accounting issues. Consequently, the notes are subject to fraudulent manipulation.

Textual Analysis Techniques

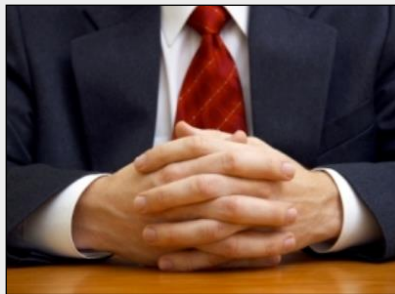
Financial Statement Text

One popular tool for assessing the readability of the notes to financial statements is the Fog Index developed by Robert Gunning in 1952. The Fog Index uses an algorithm to measure the readability of a sample of English writing. In calculating readability, the Fog Index considers average sentence length and number of words with three or more syllables ("complex" words). The calculation score reflects the number of years of formal education needed to understand the text when initially read. For example, a Fog Index score of 10 corresponds to the reading level of an average U.S. high school sophomore.

Implementation Challenges

While textual analytics can be extremely revealing and provide clues for potential frauds that might otherwise go unnoticed, fraud examiners should consider some of the challenges involved in its implementation. They should consider the costs associated with obtaining software used for text mining, and thoroughly understand the legal environment of employee rights and workplace searches. Fraud examiners who are considering adding textual analytics to their fraud detection arsenal should consult with technological and legal experts before undertaking such techniques.

Lesson 22: Conclusion



Conclusion

Businesses contain an array of complex systems, which presents a challenge for auditors. However, this complexity also provides a broad range of opportunities for the use data analysis techniques. As the business environment continues to expand and become more diverse, data analysis will evolve to keep up with this expansion.

Conclusion

As repeatedly stressed throughout this course, data analysis *alone* does not detect fraud. It is a helpful tool for a fraud examiner to use when fraud is buried deep within millions of transactions and records. Data analysis identifies anomalies in these records that might indicate the presence of fraud. If you suspect a billing scheme and a data analysis test isolates five suspicious transactions out of 50,000, you can focus your investigation in the right place.

Conclusion

Additionally, bear in mind that there are many different potential data analyses tests. The tests you need to perform depend on the fraud risks present in the organization you are investigating, what data you have available, and what fraud suspicions exist. The tests presented in this course are helpful to a wide variety of investigations, but they need to be customized to your data. In many cases, you will need to combine several data analysis techniques to narrow your scope to a manageable degree. Those who commit fraud are often very creative in hiding their misconduct. Investigators must be equally creative in their search for fraud.

You have completed this course!

Important information about the final exam:

To obtain CPE credit for this course, you must successfully achieve a score of 70% on the final exam 1) within three attempts and 2) within one year of the course purchase date. If you do not pass the exam within either three attempts or one year of the purchase date, you will not be eligible to receive CPE credit for this course.

To complete the final exam:

Close this browser window to return to the main menu, then click the Launch button in the course outline under Final Examination.